Douglas V. Guzior, Ph.D.

doug.guzior@gmail.com • guziordo.github.io • Ann Arbor, MI

Education

Ph.D., Microbiology, Genetics, & Immunology — Michigan State University | East Lansing, MI
Dissertation: Elucidating the mechanisms behind microbial bile acid conjugation and its impacts on host microbiota and physiology
Mentor: Robert A. Quinn, Ph.D

B.S., Microbiology — The Ohio State University | Columbus, OH
Thesis: Determining the impact of host-synovial fluid factors on *Staphylococcus aureus* aggregation
Mentors: Paul Stoodley, Ph.D. and Sashwati Roy, Ph.D.

Positions

Study Director
Panome Bio | St. Louis, MO — Apr 2025 - Present

Multi-Omic Analyst
Panome Bio | St. Louis, MO — May 2024 - Apr 2025

Visiting Graduate Researcher
Dr. Trevor Lawley's Lab | Wellcome Sanger Institute | Cambridge, England, UK — Feb 2023 - Mar 2023

Publications

1. Cornelius P, Mayes BA, Dannenberg AJ, Dufour PJ, Little S, **Guzior DV**, Petersen JS, Shanahan JM, Carver BJ. 2026. Evexomostat (SDX-7320), a methionine aminopeptidase type 2 inhibitor, stimulates weight loss and inhibits obesity-accelerated tumor growth. Front. Oncol. 16:1751681. doi: 10.3389/fonc.2026.1751681
2. Dean AE, **Guzior DV**, Quinn RA, Gaulke CA, Anakk S. 2025. Serum cholic acid and cecal *Faecalibaculum* increase in a male-specific manner in a murine hepatocellular carcinoma model. J. Lipid Res., 67(1), 100954. doi: 10.1016/j.jlr.2025.100954
3. González CT, Martin C, Crane M, Gutierrez K, Thomas J, Remisoski L, Okros M, Fu Y, **Guzior DV**, Finkhouse D, Bridges C, Mielke J, Querido G, Padillo L, Girgis R, McClelland M, Conrad D, Li X, Quinn RA. 2025. Microenvironmental effects of a non-antibiotic therapy for a chronic polymicrobial infection alter microbial physiology, competition, and virulence. ISME J 19:wraf125. doi: 10.1093/ismejo/wraf125
4. McGilvrey MI, Fortier B, Cooke D, Mahdi MA, Tero B, Potts CM, Kaija A, Ryzhova L, Cora C, Richardson A, **Guzior D**, Pinz I, Vary C, Koza RA, Ables G, Liaw L. 2025. Short-term dietary methionine restriction with high fat diet counteracts metabolic dysfunction in male mice. Physiol Rep 13:e70405. doi: 10.14814/phy2.70405
5. Fu Y, **Guzior DV**, Okros M, Bridges C, Rosset SL, González CT, Martin C, Karunarathne H, Watson VE, Quinn RA. 2025. Balance between bile acid conjugation and hydrolysis activity can alter outcomes of gut inflammation. Nat Commun 16:3434. doi: 10.1038/s41467-025-58649-x
6. Won TH, Arifuzzaman M, Parkhurst CN, Miranda IC, Zhang B, Hu E, Kashyap S, Letourneau J, Jin WB, Fu Y, **Guzior DV**, Quinn RA, Guo CJ, David LA, Artis D, Schroeder FC. 2025. Host metabolism balances microbial regulation of bile acid signalling. Nature 638(8049):216-224. doi: 10.1038/s41586-024-08379-9
7. Wu H[‡], **Guzior DV**[‡], Martin C, Neugebauer KA, Rzepka MM, Lumeng JC, Quinn RA, de los Campos G. 2024. Longitudinal analyses of infants' microbiome and metabolome reveal microbes and metabolites with seemingly coordinated dynamics. Commun. Biol. 7:1506. doi: 10.1038/s42003-024-07015-6 *[‡]Equal contribution towards authorship.*

8. Hansen ZA, Schilmiller AL, **Guzior DV**, Rudrik JT, Quinn R, Vasco KA, Manning SD. 2024. Shifts in the functional capacity and metabolite composition of the gut microbiome during recovery from enteric infection. Front. Cell. Infect. Microbiol. 14:1359576. doi: 10.3389/fcimb.2024.1359576

9. **Guzior DV**, Okros M, Shivel M, Armwald B, Bridges C, Fu Y, Martin C, Schilmiller A, Miller WM, Ziegler KM, Sims MD, Maddens ME, Graham SF, Hausinger RP, Quinn RA. 2024. Bile salt hydrolase/acyltransferase activity shapes bile acid diversity. Nature 626:852–858. doi: 10.1038/s41586-024-07017-8

10. Chargo NJ, Neugebauer K, **Guzior DV**, Quinn RA, Parameswaran N, McCabe LR. 2024. Glucocorticoid-induced osteoporosis is prevented by dietary prune in female mice. Front. Cell. Dev. Biol. 11:1324649. doi: 10.3389/fcell.2023.1324649

11. Martin C, **Guzior DV**, Gonzalez CT, Okros M, Mielke J, Padillo L, Thomas R, McClelland M, Conrad D, Widder S, Quinn RA. 2023. Longitudinal microbial and molecular dynamics in the cystic fibrosis lung after Elexacaftor-Tezacaftor-Ivacaftor therapy. Respir. Res. 24:317. doi: 10.1186/s12931-023-02630-z

12. Quinn RA, Martin C, **Guzior DV**. 2023. A multi-omic trip through the human gut. Nat. Metab. 5:720–721. doi: 10.1038/s42255-023-00773-3

13. Neugebauer KA, Okros M, **Guzior DV**, Feiner J, Chargo NJ, Rzepka M, Schilmiller AL, O'Reilly S, Jones AD, Watson VE, Luyendyk JP, McCabe LR, Quinn RA. 2022. Baat gene knockout alters post-natal development, the gut microbiome, and reveals unusual bile acid conjugates in mice. J. Lipid Res. 63:100297. doi: 10.1016/j.jlr.2022.100297

14. Ghuneim L-AJ, Raghuvanshi R, Neugebauer KA, **Guzior DV**, Martin C, Schena B, Feiner JM, Castillo-Bahena A, Mielke J, McClelland M, Conrad D, Klapper I, Zhang T, Quinn RA. 2022. Complex and unexpected outcomes of antibiotic therapy against a polymicrobial infection. ISME J 16:2065–2075. doi: 10.1038/s41396-022-01252-5

15. Wurster JI, Peterson RL, Brown CE, Penumutchu S, **Guzior DV**, Neugebauer KA, Sano WH, Sebastian MM, Quinn RA, Belenky P. 2021. Streptozotocin-induced hyperglycemia alters the cecal metabolome and exacerbates antibiotic-induced dysbiosis. Cell. Rep. 37:110113. doi: 10.1016/j.celrep.2021.110113

16. Sosinski LM, Martin C, Neugebauer KA, Ghuneim L-AJ, **Guzior DV**, Castillo-Bahena A, Mielke J, Thomas R, McClelland M, Conrad D, Quinn RA. 2022. A restructuring of microbiome niche space is associated with Elexacaftor-Tezacaftor-Ivacaftor therapy in the cystic fibrosis lung. J. Cyst. Fibros. 21:996–1005. doi: 10.1016/j.jcf.2021.11.003

17. **Guzior DV**, Quinn RA. 2021. Review: microbial transformations of human bile acids. Microbiome 9:140. doi: 10.1186/s40168-021-01101-1

18. Pestrak MJ, Gupta TT, Dusane DH, **Guzior DV**, Staats A, Harro J, Horswill A, Stoodley P. 2020. Investigation of synovial fluid induced *Staphylococcus aureus* aggregate development and its impact on surface attachment and biofilm formation. PLoS ONE 15:e0231791. doi: 10.1371/journal.pone.0231791

Conference Presentations

Invited Talks

Guzior DV. The new dogma of microbial bile acid metabolism. Corewell Health Metabolomics Symposium; August 2024; Grand Rapids, MI. *Early Career Invited Speaker.*

Selected Talks

Guzior DV, Wu H, Martin C, Rzepka M, Neugebauer KA, Lumeng J, de los Campos G, Quinn RA. Developmental dynamics of the infant fecal metabolome. MANA 2023; October 2023; Columbia, MO.

Guzior DV, Quinn RA. Bile salt hydrolase aminoacyltransferase activity from gut anaerobes expands the human bile acid pool. Anaerobe 2023; July 2023; Cardiff, Wales, UK. *Awarded Best Offered Talk.*

Guzior DV, Quinn RA. Reversing Dogma: Bile Salt Hydrolase. Anaerobe Congress; July 2022; Seattle, WA, USA.

Awards & Fellowships

2024	Dissertation Completion Fellowship, The Graduate School, Michigan State University
2023	Graduate School Travel Fellowship, The Graduate School, Michigan State University
2023	Ralph Evans Award, Department of Microbiology, Genetics, & Immunology, Michigan State University
2023 Summer **2023 Spring** **2022 Autumn**	Outstanding Scholar Fellowship, The Graduate School, Michigan State University
2021	Philipp and Vera Gerhardt Award, Department of Microbiology, Genetics, & Immunology, Michigan State University

Professional Memberships

Metabolomics Association of North America

Anaerobe Society of the Americas

American Society for Microbiology

Microbiology Society

Leadership & Service

Leadership

2022-2024	Graduate Committee Representative, Department of Microbiology, Genetics, & Immunology, Michigan State University

Service

2015-2023	Annual Fundraiser and Shavee, St. Baldrick's Foundation (over $8,000 raised)
2022-2024	Graduate Student Mentor, Graduate Recruitment Initiative Team, Michigan State University
2020	Volunteer, Science Debate - MSU Chapter
2020	Donor, National Marrow Donor Program

Skills & Expertise

- Targeted and untargeted metabolomics
- Heterologous gene expression, mutagenesis
- Enzyme kinetic characterization
- Leadership and mentorship
- Teamwork and collaboration
- Microbiome analysis
- Multi-Omics analysis
- R, Python
- Written and oral communication
- Statistics and data visualization